362 Project Notes

Implementation

Menu

SSTV Menu UI

Requirements

- Peripherals with 4 DIFFERENT architectures - SPI, UART, GPIO
- Only one main ‘computer’

Nomenclature

- SDRs: Software Defined Radios

- DSM: Delta–Sigma Modulator

Oversampling analog-to-digital converter (ADC)
Noise Shaping <- Quantization noise, pushing most of it out of the signal band and into higher frequencies

- I/Q: (In-phase) and Q (Quadrature)

Complete mathematical representation of how a carrier wave is modulated

Amplitude: $|I + jQ| = \sqrt{I^2 + Q^2}$

Phase: $\phi = \arctan(Q/I)$

Frequency: Changes in phase over time

Documentation

RP2350

Main SDK Repository

https://github.com/raspberrypi/pico-sdk

Hardware APIs

https://www.raspberrypi.com/documentation/pico-sdk/hardware.html

NESDR SMArt v5

https://www.nooelec.com/store/downloads/dl/file/id/111/product/249/nesdr_smart_rtl_sdr_v5_datasheet_revision_1.pdf?srsltid=AfmBOopmAHBOuylUt8R4TPNeTvh07Hk6CnjebHT-qKU-Z8Lz9kJnQDzm

Coding Snippets